SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks: 25**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I M.Nivetha (192421307) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.Nivetha

Annexure A

Short Answer Test

1. With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme. (5 Marks)

Block Diagram

```
   Data Collection
        |
        ▼
  Data Preprocessing
(Cleaning, Normalization)
        |
        ▼
  Feature Engineering
(Feature Selection/Extraction)
        |
        ▼
  Machine Learning Model
(Training Algorithm)
        |
        ▼
   Model Evaluation
(Accuracy, Precision, Recall)
        |
        ▼
  Prediction  /  Deployment
(Final Output)
```

Blocks

1. **Data Collection** – Gather data.
2. **Data Preprocessing** – Clean and prepare data.
3. **Feature Engineering** – Select important features.
4. **Model Training** – Train the ML algorithm.
5. **Model Evaluation** – Check model performance.
6. **Prediction** – Predict output for new data.

Supervised Learning

- Uses **labeled data** (input + output).
- **Example:** Email spam detection or house price prediction.

Unsupervised Learning

- Uses **unlabeled data**.
- **Example:** Customer segmentation using clustering.

Result: Supervised learning predicts known outputs, while unsupervised learning discovers hidden patterns in data.

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data. Explain the choice of model, data preprocessing, and feature selection techniques.
(a) Describe the steps involved in training, validating, and evaluating the model.
(b) Explain how the system can assist in early warning and emergency response.
(5 Marks)

System Design

- **Data:** Historical and real-time weather, rainfall, river level, and seismic data.
- **Model:** Random Forest or Neural Network for prediction.
- **Preprocessing:** Remove missing values, normalize data, and clean noise.
- **Feature Selection:** Rainfall, temperature, river level, earthquake magnitude, and location.

(a) Training, Validation, and Evaluation

1. Collect and preprocess data.
2. Split into **training** and **validation** datasets.
3. Train the ML model.
4. Validate and tune the model.
5. Evaluate using **Accuracy, Precision, Recall,** and **F1-score**.

(b) Early Warning and Emergency Response

- Predict floods and earthquakes in advance.
- Send early warning alerts to people.
- Help authorities plan evacuation and allocate rescue resources. ● Reduce loss of life and property.

Result: The ML system analyzes historical and real-time data to accurately predict disasters and support timely emergency response.

3. In the context of Find-S algorithm, consider the following dataset used to determine whether a day is suitable for playing tennis based on attributes such as Outlook, Temperature, Humidity, and Wind. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show the step-by-step updates of the hypothesis after each positive training example and ignore all negative examples. Finally, provide the final hypothesis. (5 Marks)

Dataset:

Example	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

Initialize Hypothesis

H_0 = ⟨Ø, Ø, Ø, Ø⟩

Ignore all **negative examples (No)**.

Positive Example	Instance	Updated Hypothesis
Example 3	(Overcast, Hot, High, Weak)	**H_1 = ⟨Overcast, Hot, High, Weak⟩**
Example 4	(Rainy, Mild, High, Weak)	**H_2 = ⟨?, ?, High, Weak⟩**
Example 5	(Rainy, Cool, Normal, Weak)	**H_3 = ⟨?, ?, ?, Weak⟩**
Example 7	(Overcast, Mild, Normal, Strong)	**H_4 = ⟨?, ?, ?, ?⟩**

Final Maximally Specific Hypothesis

H=⟨?, ?, ?, ?⟩

The final hypothesis is ⟨**?, ?, ?, ?**⟩, meaning none of the attributes consistently describe all positive training examples, so all attributes are generalized.

4. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a person will enjoy a sport based on attributes such as Weather, Temperature, Humidity, and Wind. Using the Candidate Elimination algorithm, determine the version space by computing both the specific (S) and general (G) boundaries. Show the step-by-step updates after processing each training example.

(5 Marks)

Dataset:

Example	Weather	Temperature	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Weak	Yes
2	Sunny	Warm	High	Strong	Yes
3	Rainy	Cold	High	Weak	No
4	Sunny	Cold	Normal	Weak	Yes
5	Rainy	Warm	Normal	Strong	No

nitialize

- **Specific Boundary (S)** = ⟨Ø, Ø, Ø, Ø⟩
- **General Boundary (G)** = ⟨?, ?, ?, ?⟩

Step-by-Step Updates

Example	Clas s	Specific Boundary (S)	General Boundary (G)
1 (Sunny, Warm, Normal, Weak)	Yes	⟨Sunny, Warm, Normal, Weak⟩	⟨?, ?, ?, ?⟩
2 (Sunny, Warm, High, Strong)	Yes	⟨Sunny, Warm, ?, ?⟩	⟨?, ?, ?, ?⟩
3 (Rainy, Cold, High, Weak)	No	⟨Sunny, Warm, ?, ?⟩	{⟨Sunny, ?, ?, ?⟩, ⟨?, Warm, ?, ?⟩}
4 (Sunny, Cold, Normal, Weak)	Yes	⟨Sunny, ?, ?, ?⟩	{⟨Sunny, ?, ?, ?⟩}
5 (Rainy, Warm, Normal, Strong)	No	⟨Sunny, ?, ?, ?⟩	{⟨Sunny, ?, ?, ?⟩}

Final Version Space

- **Specific Boundary (S): ⟨Sunny, ?, ?, ?⟩**
- **General Boundary (G): {⟨Sunny, ?, ?, ?⟩}**

Result

The final hypothesis is ⟨**Sunny, ?, ?, ?**⟩, indicating that a person is predicted to enjoy the sport whenever the **Weather is Sunny**, regardless of the other attributes.

5. **In the context of the Find-S algorithm, consider the dataset used to determine whether a customer will buy a product based on attributes such as Age, Income, Student, and Credit Rating. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show step-by-step updates after each positive example and ignore negative examples. Finally, provide the final hypothesis.** (5 Marks)

Dataset:

Example	Age	Income	Student	Credit Rating	Buys Product
1	Young	High	No	Fair	No
2	Young	High	No	Excellent	No
3	Middle	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes

nitialize

H_0 = ⟨Ø, Ø, Ø, Ø⟩

Ignore **negative examples (No)**.

Positive Example	Instance	Updated Hypothesis
Example 3	(Middle, High, No, Fair)	**H_1 = ⟨Middle, High, No, Fair⟩**
Example 4	(Senior, Medium, No, Fair)	**H_2 = ⟨?, ?, No, Fair⟩**

Final Maximally Specific Hypothesis H=⟨?, ?, No, Fair⟩H=⟨?, ?, No, Fair⟩

Result: The customer is likely to buy the product when **Student = No** and **Credit Rating = Fair**.

6. In the context of the **Find-S algorithm**, consider a dataset used to determine whether a person will go for a workout based on attributes such as Weather, Temperature, Time, and Mood. Apply the Find-S algorithm and determine the maximally specific hypothesis. Show the hypothesis updates after each positive example only. (5 Marks)

Dataset:

Example	Weather	Temperature	Time	Mood	Workout
1	Sunny	Hot	Morning	Tired	No
2	Rainy	Mild	Evening	Happy	Yes
3	Sunny	Mild	Evening	Happy	Yes
4	Cloudy	Cool	Morning	Happy	Yes

Initialize

H_0 = ⟨Ø, Ø, Ø, Ø⟩

Ignore **negative examples (No)**.

Positive Example	**Instance**		**Updated Hypothesis**
Example 2	(Rainy, Happy)	Mild,	Evening, **H_1 = ⟨Rainy, Mild, Evening, Happy⟩**
Example 3	(Sunny, Happy)	Mild,	Evening, **H_2 = ⟨?, Mild, Evening, Happy⟩**
Example 4	(Cloudy, Happy)	Cool,	Morning, **H_3 = ⟨?, ?, ?, Happy⟩**

Final Maximally Specific Hypothesis H=⟨?, ?, ?, Happy⟩H=⟨?, ?, ?, Happy⟩

Result: The person is likely to work out when the **Mood = Happy**, regardless of the other attributes.

7. In the context of the **Find-S algorithm**, consider a dataset used to determine whether a loan should be approved based on attributes such as Employment Status, Salary Level, Credit Score, and Loan History. Apply the Find-S algorithm to derive the most specific hypothesis consistent with the positive examples. (5 Marks) **Dataset:**

Example	Employment	Salary	Credit Score	Loan History	Loan Approved
1	Employed	High	Good	Clean	Yes
2	Employed	Medium	Good	Clean	Yes
3	Unemployed	Low	Poor	Bad	No
4	Employed	Medium	Excellent	Clean	Yes

Initialize

H_0 = ⟨Ø, Ø, Ø, Ø⟩

Ignore **negative examples (No)**.

Positive Example	Instance	Updated Hypothesis
Example 1	(Employed, High, Good, Clean)	**H_1 = ⟨Employed, High, Good, Clean⟩**
Example 2	(Employed, Medium, Good, Clean)	**H_2 = ⟨Employed, ?, Good, Clean⟩**
Example 4	(Employed, Medium, Excellent, Clean)	**H_3 = ⟨Employed, ?, ?, Clean⟩**

Final Maximally Specific Hypothesis

H=⟨Employed, ?, ?, Clean⟩H=⟨Employed, ?, ?, Clean⟩

Result: Loan is approved for **Employed** applicants with a **Clean Loan History**, regardless of salary and credit score.

8. In the context of the **Candidate Elimination algorithm**, consider a dataset used to classify whether a customer will subscribe to a service based on attributes such as Age Group, Income Level, Internet Usage, and Device Type. Determine the version space by computing the **specific (S)** and **general (G)** boundaries. Show step-by-step updates after each training example. (5 Marks)

Dataset:

Example	Age Group	Income Level	Internet Usage	Device Type	Subscribe
1	Young	High	Frequent	Mobile	Yes
2	Young	High	Occasional	Mobile	Yes
3	Middle	Low	Frequent	Desktop	No
4	Young	Medium	Frequent	Mobile	Yes
5	Senior	High	Occasional	Desktop	No

Initialize

- **Specific Boundary (S)** = ⟨Ø, Ø, Ø, Ø⟩
- **General Boundary (G)** = ⟨?, ?, ?, ?⟩

Example Clas Specific General Boundary (G) s Boundary (S)

1 (Young, High, Frequent, Mobile)	Yes	⟨Young, High, Frequent, Mobile⟩	⟨?, ?, ?, ?⟩
2 (Young, High, Occasional, Mobile)	Yes	⟨Young, High, ?, Mobile⟩	⟨?, ?, ?, ?⟩
3 (Middle, Low, Frequent, Desktop)	No	⟨Young, High, ?, Mobile⟩	{⟨Young, ?, ?, ?⟩, ⟨?, High, ?, ?⟩, ⟨?, ?, ?, Mobile⟩}
4 (Young, Medium, Frequent, Mobile)	Yes	⟨Young, ?, ?, Mobile⟩	{⟨Young, ?, ?, ?⟩, ⟨?, ?, ?, Mobile⟩}
5 (Senior, High, Occasional, Desktop)	No	⟨Young, ?, ?, Mobile⟩	{⟨Young, ?, ?, ?⟩, ⟨?, ?, ?, Mobile⟩}

Final Version Space

- **Specific Boundary (S): ⟨Young, ?, ?, Mobile⟩**
- **General Boundary (G): {⟨Young, ?, ?, ?⟩, ⟨?, ?, ?, Mobile⟩}**

Result: The final version space indicates that customers are likely to subscribe if they are **Young** and/or use a **Mobile**device.

9. In the context of the **Candidate Elimination algorithm**, consider a dataset used to classify whether a patient follows a healthy lifestyle based on attributes such as Diet, Exercise, Sleep, and Stress Level. Apply the algorithm to compute the **version space (S and G boundaries)** and show the updates after each example. (5 Marks) **Dataset:**

Example	Diet	Exercise	Sleep	Stress Level	Healthy Lifestyle
1	Balanced	Regular	Adequate	Low	Yes
2	Balanced	Regular	Poor	Medium	Yes
3	Junk	None	Poor	High	No
4	Balanced	Moderate	Adequate	Low	Yes
5	Junk	Regular	Adequate	High	No

Initialize

- **Specific Boundary (S)** = ⟨∅, ∅, ∅, ∅⟩
- **General Boundary (G)** = ⟨?, ?, ?, ?⟩

Example	Class	Specific Boundary (S)	General Boundary (G)
1 (Balanced, Regular, Adequate, Low)	Yes	⟨Balanced, Regular, Adequate, Low⟩	⟨?, ?, ?, ?⟩

2 (Balanced, Regular, Poor, Medium)	Yes	⟨Balanced, Regular, ?, ?⟩	⟨?, ?, ?, ?⟩
3 (Junk, None, Poor, High)	No	⟨Balanced, Regular, ?, ?⟩	{⟨Balanced, ?, ?, ?⟩, ⟨?, Regular, ?, ?⟩}
4 (Balanced, Moderate, Adequate, Low)	Yes	⟨Balanced, ?, ?, ?⟩	{⟨Balanced, ?, ?, ?⟩}
5 (Junk, Regular, Adequate, High)	No	⟨Balanced, ?, ?, ?⟩	{⟨Balanced, ?, ?, ?⟩}

Final Version Space

- **Specific Boundary (S): ⟨Balanced, ?, ?, ?⟩**
- **General Boundary (G): {⟨Balanced, ?, ?, ?⟩}**

Result: A patient is predicted to follow a heathy lifestyle if the **Diet is Balanced**, regardless of the other attributes.

10. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions. (5 Marks)

System Design

- **Data:** Age, Blood Pressure, Glucose Level, BMI, Lifestyle, Medical History.
- **Algorithm:** Decision Tree, Random Forest, or Logistic Regression.
- **Reason:** These algorithms provide good accuracy and are suitable for disease prediction.

Steps

1. **Data Collection** – Collect patient records.
2. **Data Preprocessing** – Remove missing values, normalize data, and encode categorical values.
3. **Feature Selection** – Select important features such as age, blood pressure, glucose, and BMI.
4. **Model Training** – Train the selected ML algorithm using training data.
5. **Model Evaluation** – Test using Accuracy, Precision, Recall, and F1-score.

Result

The ML system predicts diseases such as **heart disease** or **diabetes** accurately, helping doctors with early diagnosis and treatment.

11. Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples. (5 Marks)

Diagram

Data Collection

↓

Data Preprocessing

↓

Feature Engineering

↓

Model Training

↓

Model Evaluation

↓

Deployment

Stages

- **Data Collection:** Gather data from various sources.
- **Data Preprocessing:** Clean, normalize, and remove missing values.
- **Feature Engineering:** Select or create useful features.
- **Model Training:** Train the ML algorithm using training data.
- **Model Evaluation:** Measure performance using accuracy, precision, recall, etc.
- **Deployment:** Use the trained model for real-world predictions.

Training Error vs Generalization Error

Training Error	Generalization Error
Error on training data	Error on unseen test data

Usually lower	
	Indicates real-world performance
Example: 95% training accuracy	Example: 90% test accuracy

Result: A machine learning pipeline converts raw data into accurate predictions through sequential stages.

12. Design a machine learning model for spam email classification using labeled data. (5 Marks)
(a) Explain the choice of algorithm (e.g., Naïve Bayes, SVM, or Logistic Regression).
(b) Describe the preprocessing steps such as tokenization, stop-word removal, and vectorization.
(c) Discuss evaluation metrics such as accuracy, precision, recall, and F1-score.

a) Choice of Algorithm

- **Naïve Bayes** is commonly used because it is simple, fast, and works well for text classification.
- (SVM or Logistic Regression can also be used.)

(b) Preprocessing Steps

1. Tokenization (split text into words).
2. Remove stop words (e.g., *the, is, and*).
3. Convert text into numerical vectors using **TF-IDF** or **Bag of Words**.

(c) Evaluation Metrics

- **Accuracy:** Overall correct predictions.
- **Precision:** Correctly identified spam emails.
- **Recall:** Ability to detect all spam emails.
- **F1-Score:** Harmonic mean of precision and recall.

Result: The ML model classifies emails as **Spam** or **Not Spam** with high accuracy using labeled training data.

13. In the context of the Candidate-Elimination Algorithm, consider a dataset with attributes: (5 Marks)
Sky, AirTemp, Humidity, Wind, Water, Forecast and target concept *EnjoySport*.
(a) Explain the initialization of S (specific boundary) and G (general boundary).

(b) Show how S and G change with each training example.
(c) Compare Candidate-Elimination with the Find-S algorithm in terms of generalization capability.

(a) Initialization

- **Specific Boundary (S):** ⟨Ø, Ø, Ø, Ø, Ø, Ø⟩ (most specific)
- **General Boundary (G):** ⟨?, ?, ?, ?, ?, ?⟩ (most general)

(b) Updating S and G

- **Positive Example:** Generalize **S** to include the example.
- **Negative Example:** Specialize **G** to exclude the example.
- Continue updating until all training examples are processed.

(c) Candidate-Elimination vs Find-S

Candidate-Elimination	Find-S
Maintains both **S** and **G** boundaries	Maintains only **S**
Uses positive and negative examples	Uses only positive examples
Finds the complete version space	Finds one maximally specific hypothesis

Result: Candidate-Elimination is more powerful because it considers both positive and negative examples.

14. Develop a machine learning system for predicting house prices using features like location, size, number of rooms, and amenities. (5 Marks) (a) Justify the choice of regression model (Linear Regression, Decision Tree, or Random Forest).
(b) Explain feature scaling and handling missing data.
(c) Describe how overfitting can be detected and avoided.

(a) Choice of Model

- **Random Forest Regression** – High accuracy and handles complex relationships.
- (Linear Regression is suitable for simple linear data.)

(b) Feature Scaling & Missing Data

- **Feature Scaling:** Normalize or standardize numerical features.
- **Missing Data:** Replace missing values with mean, median, or mode.

(c) Overfitting

- **Detection:** High training accuracy but low test accuracy.
- **Prevention:** Cross-validation, pruning, regularization, and using more training data.

Result: The model predicts house prices accurately using features such as location, size, and amenities.

15. Explain the concept of clustering in unsupervised learning. (5 Marks)
(a) Describe the working of the K-Means algorithm with a step-by-step example.
(b) Explain how the value of *K* is chosen using the Elbow Method.
(c) Compare K-Means with Hierarchical Clustering.

Clustering

Clustering groups similar data points without using labeled data.

(a) K-Means Algorithm

1. Choose **K** clusters.
2. Select initial centroids.
3. Assign each data point to the nearest centroid.
4. Recalculate centroids.
5. Repeat until centroids do not change.

Example: Group customers into different shopping behavior categories.

(b) Elbow Method

- Plot **K** vs **WCSS (Within-Cluster Sum of Squares)**.
- Choose the **K** value where the graph forms an **elbow**.

(c) K-Means vs Hierarchical Clustering

K-Means	Hierarchical
Requires K value	K not required initially
Faster	Slower

Suitable for large datasets	Suitable for small datasets

Result: K-Means is simple and efficient for clustering, while Hierarchical Clustering provides a tree-like structure of clusters.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand